source ..... make noise · soundhack
rev 2026.10

spectraphon _travelers guide _for mastering alien alchemy

2 sides | **4** modes each | **16** arrays per side

› then noon = 12 o'clock tags = the panel drawing

dual spectral oscillator · each side: sam (input drives harmonics) or sao (plays stored arrays) · noise and chaos modes · by tom erbe

01 what it is

sp_01

two sides	A and B, mirror images · each is an oscillator in its own mode
a spectrum	a set of harmonic levels · not a recording · plays at any pitch
like a complex osc	freq, partials, fm bus, follow/sync and sine work the same in every mode

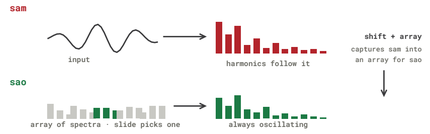
02 modes

sp_02 · fig.01 · mode button

sam · off	the input sets harmonic levels · silent without input
sao · lit	always oscillates · plays spectra from a stored array
noise · chaos	press on past sao (fw sp52+) · the centre LEDs show which: noise 01, chaos 10
mix them	each side has its own mode · fm one with the other either way

__fig.01 : sam and sao

mode off · mode lit



10 first flight

sp_10 · first sound

1. side A: mode off (sam)
2. a voice or drum loop into a-in · knob to the tick
3. partials full · listen to odd
4. turn slide then focus
5. play 1v/oct like any vco
6. [shift] + [array] to capture, then press mode for sao

14 tips

sp_14 · manual p.34 – 35

odd vs even	echo only the evens · drone the odds, sequence evens with in
stereo ripple	pan odd and even apart, then move partials
cross-patch	side A's follower into its slide · side B's random into A's focus
trigger voice	sam: trigger into the input, follower into focus
sam → sao	unpatch the input: in sao it detunes the evens

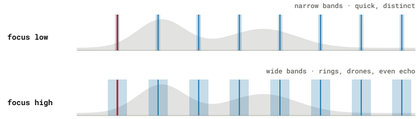
03 sam

sp_03 · spectral amplitude modulation

a-in	input level · ticks mark modular and line level
slide	where the input's first harmonic sits · lower = brighter output
focus	band width · low = sharp, quick · high = rings, drones, even echo
sub/cv	envelope follower of the input · try it into slide or focus

__fig.05 : slide · focus (sam)

what the bands hear



grey = the input · red line = first harmonic, set by slide
slide lower than the input's pitch = brighter output

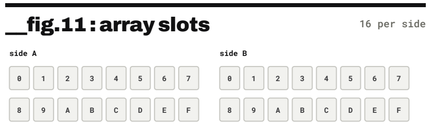
04 sao

sp_04 · spectral array oscillation

slide	picks the spectrum: first (ccw) to last (cw) in the array
focus	cw = compress (brighter, even) · ccw = expand (darker, punchier)
[shift] 2 s	planar style: focus = fine, slide = coarse
clk	steps through the array · adds to slide and focus
a-in	detunes the even harmonics up to an octave · cv = fm for evens only

__fig.11 : array slots

16 per side



files SPECA000.WAV … SPECB015.WAV on the microSD · empty slot = default array
hold shift (this side) + shift (other side) to pick · centre LEDs show it in binary

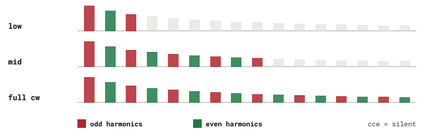
05 shared controls

sp_05 · every mode

freq + fine	core pitch with 1v/oct · sine always plays it, even under fm
partials	silent → low → high harmonics · alternates odd and even · a timbre vca
odd even	unpatch even and odd carries both
sub/cv	sao: sub osc (A saw, B saturated sine)

__fig.04 : partials

low → high · odd / even alternate



06 fm bus · follow · sync

sp_06

index A index B	internal fm from the other side · both ways at once, no feedback
[follow/sync]	lit = follow: B pitch becomes an offset from A (noon = 0)
flashing	sync: follow + B hard-synced to A · strong sweeps
beacon	green = octave, fifth, fourth · red = third, sixth
tip	turn follow on before tuning B

__fig.07 : tuning beacon

A : B

2 : 1 octave · 3 : 2 fifth · 4 : 3 fourth · 5 : 4 major third · 5 : 3 sixth

ratios of A to B, just intonation · not equal temperament
turn follow on first, then tune B until the beacon lights

07 shift layer

sp_07 · hold shift

[shift] + [array]	sam: start / stop array capture · other modes: low frequency (shift stays lit)
[shift] + [cv]	sub/cv: sub · stepped random · smooth random · triangle · ramp
[shift] + other	shift pick array 0 – 15 for this side (binary on the centre LEDs)
… hold 2 s	delete that array, back to default
[shift]	tap = manual clock · tap tempo for the LFOs

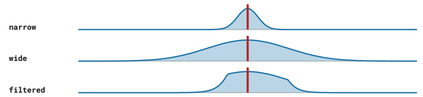
08 noise · chaos

sp_08 · fw sp52+

noise	two sines + noise · partials = sideband width · slide lp, focus hp
chaos	each sine gets a partner · focus = ratio · partials = cross-mod · slide = feedback
a-in	detunes the pair up to an octave, as in sao

__fig.10 : noise mode

sidebands



red = the sine · partials = sideband width · slide = lowpass · focus = highpass

11 capturing arrays

sp_11 · fig.02

before — partials full
fm bus off · partials full cw and unmodulated

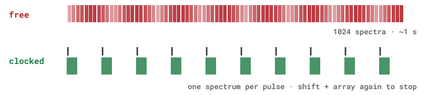
free — no clock 3 s
1024 spectra in about 1 s

clocked — clock in
one spectrum per pulse · shift + array again stops it

tip — keep it moving
static or silent parts become dead spots on slide

__fig.02 : capture

sam · shift + array



13 sub/cv · clocking

sp_13 · shift + cv · manual p.31

1	sub osc (sao) · envelope follower (sam) · sub also in sam (sp38+)
2 · 3	stepped random · smooth random
4 · 5	triangle lfo · ramp lfo
clk + [shift]	in modes 2 – 5 these clock the cv, not the array
no clock	tap [shift] for tempo · slowest one tap per 3 s

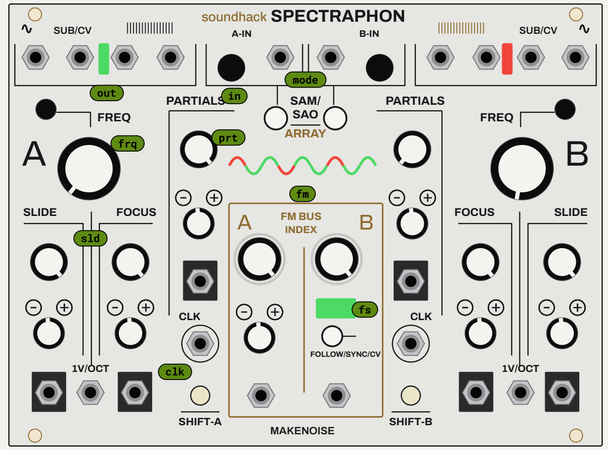
09 specs

sp_09

size	34 hp
power	230 mA +12 V · 55 mA –12 V
arrays	16 per side on microSD · SPECA000.WAV …
firmware	sp_.dat on the card · power on holding mode B

__fig.03 : the panel

numbers match the legend



out	outputs · sine · sub/cv · odd · even
in	a-in · b-in · input level / even detune
mode	sam/sao/array · mode · shift + = capture / low freq
frq	freq · fine · core pitch + 1v/oct
prt	partials · harmonic amount
sld	slide · focus · shape or pick the spectrum
fm	fm bus index · A ← B · B ← A
fs	follow/sync · B follows or syncs to A
clk	clk · shift · step array · second layer

12 reset

sp_12

modes	power on holding both [shift] buttons
one array	delete its file from the card, or hold both shifts 2 s

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel drawing after the make noise panel
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